



# Department of Computer Science University of Management and Technology

## Week 2 Status Report: Topology & Infrastructure Setup

**Project Title:** Campus Network (CCP)

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**Date:** May 20, 2026.

**Status:** Blocked / Troubleshooting Phase

### Work Completed This Week:

- **Topology Mapping:** Successfully mapped the core physical architecture in GNS3 (Student, Staff, and Guest networks).
- **Environment Troubleshooting:** Encountered severe local HAXM/virtualization errors that prevented the use of standard heavy switch images.
- **Hardware Workaround Execution:** Researched and deployed a c3745 router outfitted with an NM-16ESW hardware module to act as a core Layer 2 EtherSwitch, successfully bypassing the virtualization blockers.
- **Physical Cabling:** Established the physical link connections between the core router, the EtherSwitch, and all end devices (VPCS).

### Current Blockers

**Emulator Instability:** The team is currently blocked from executing the CLI configurations (VLANs, inter-VLAN routing, and Captive Portal ACLs) due to extensive software instability. The Dynamips emulator experienced persistent console freezing and crashes during legacy VLAN database configurations, requiring significant troubleshooting time that halted configuration progress.

### **Next Steps (Week 3)**

- Execute the pending Layer 2 and Layer 3 CLI configurations on the stabilized topology.
- Assign static IP addresses to all end devices.
- Begin the Ubuntu server integration for the Captive Portal once routing is fully operational.